

Molecular Shapes (VSEPR)

Electron domains, shapes, bond angles and polarity

Name _____ Class _____ Date _____

ELECTRON DOMAINS

2 linear 180° · 3 trigonal planar 120° · 4 tetrahedral 109.5°

5 trigonal bipyramidal 90° and 120° · 6 octahedral 90°

A double or triple bond counts as one domain

Draw the Lewis structure first, then count domains on the central atom.

A. Understanding

1. State the main idea of VSEPR theory.

2. Explain the difference between the electron-domain geometry and the molecular shape, using water as an example.

B. Predicting shapes

3. For each molecule, give the number of electron domains and lone pairs on the central atom, the molecular shape and the approximate bond angle: (a) BCl_3 (b) H_2S (c) PCl_3 (d) CS_2 (e) SiH_4

4. Predict the shape around the carbon atom in formaldehyde, CH_2O , and the approximate H-C-H bond angle.

5. Predict the shape of the ammonium ion, NH_4^+ .

C. Bond angles

6. The bond angles in CH_4 , NH_3 and H_2O are 109.5° , about 107° and about 104.5° . Explain this trend.

D. Polarity

7. State whether each molecule is polar or nonpolar, with a reason: (a) CO_2 (b) H_2O (c) CCl_4 (d) NH_3

E. Challenge

8. XeF_4 has six electron domains on xenon, two of which are lone pairs. Predict its molecular shape and explain where the lone pairs go.

9. SF_4 has five electron domains on sulfur, one of which is a lone pair. Name its shape.